Cloud Computing Architecture

Assignment 3 – Group 32

Lab Section: Friday 4:30pm – 6:30pm

Serverless/Event-driven Architectural Design Report

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I. Introduction

In the digital world, there are many new technology advancements that play a crucial role in business operations to be success. Among these advancements, cloud computing is an essential tool, with AWS (Amazon Web Services) enabling businesses to ensure their web applications are scalable, and highly efficient.

Additionally, Cloud computing provides businesses with an efficient way to deploy applications by offering scalable resources, reducing infrastructure costs, ensuring fast experience and reliable access for users. One of the key advantages of AWS is Serverless architecture operates based on predefined events, ensuring that the website remains highly responsive and scalable. By adopting serverless technology, the focus shifts toward building an efficient architecture without the overhead of server management.

Furthermore, the serverless architecture for the Photo Album application enables easy uploads of images and videos by using lambda function, ensuring an efficient user experience to upload files in multiple formats, such as JPG, PNG, MP4, and more, directly through the application interface.

II. Architecture Overview

A. Architectural Designs

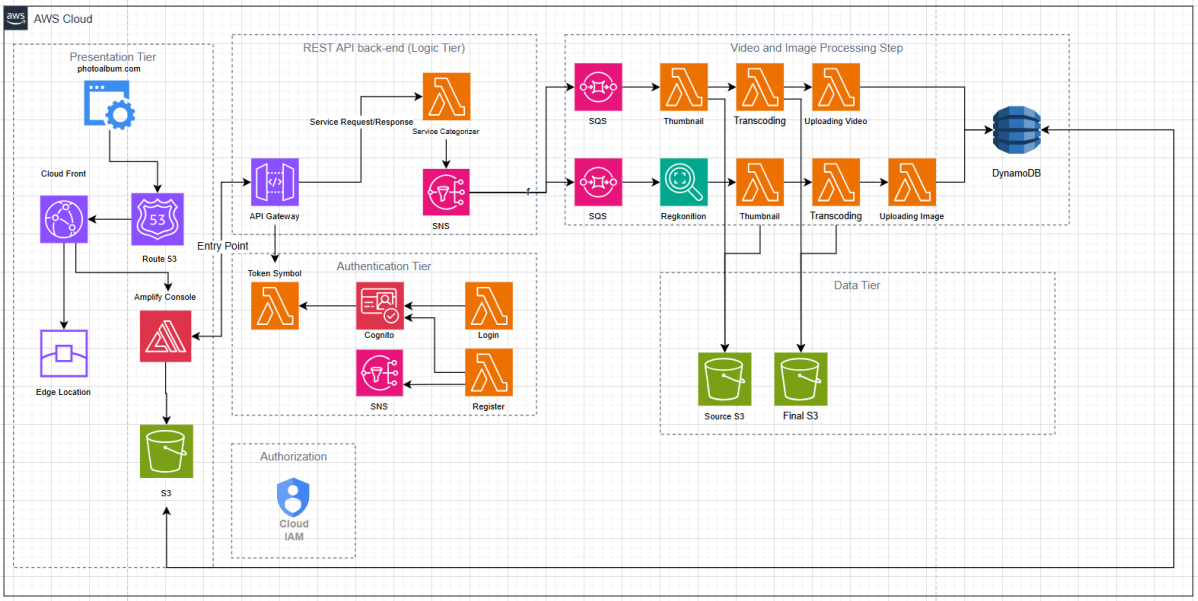


Figure 1 Architectural Designs

B. Collaboration Diagrams

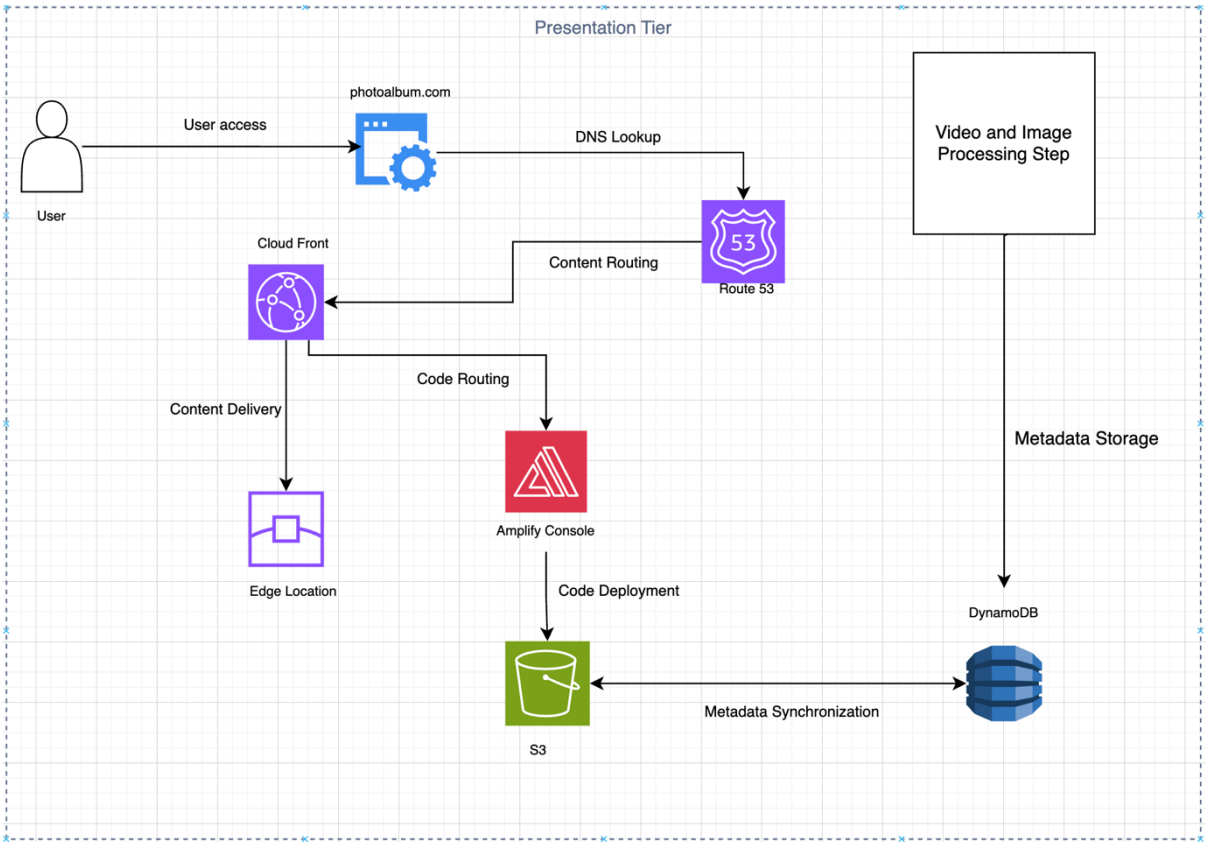


Figure 2 Presentation Tier

In this section, the Presentation Tier is the preparation of the web server of the photo album ensures efficient is the ensure the delivery of front-end website content to the users accessing the photo album website. When a user visits the photo album website (photoalbum.com), AWS Route 53 directs the request to CloudFront. CloudFront checks if the website content is already stored at the closest location to the user. If it is, the content is delivered immediately. If not, CloudFront gets the latest content from the S3 bucket managed by AWS Amplify, saves it locally, and sends it to the user. Additionally, DynamoDB works with S3 to store and manage metadata for media files, such as image and video details. When a user uploads or accesses content, the application quickly retrieves this data from DynamoDB to provide dynamic information. This setup ensures quick delivered content. This approach guarantees fast and efficient delivery with automatic updates to the user.

There are 7 component that contribute to this tier

- The explanation of the component function:

- Route 53: This is AWS domain name service which direct user traffic to the correct domain’s DNS service (photoalbum.com) and route traffic to Cloud Front and Amplify Console.
- Cloud Front: This is the Content Delivery Network that stores the website's source code at various locations around the world, enabling efficient delivery of content from the server closest to the user.
- Amplify Console: AWS Amplify handles the front-end website code deployment. When updates are made such as changing in layout or image on the website, it automatically forwards them to the S3 bucket, so CloudFront can access the latest version when needed.
- Edge Location: This is a data center worldwide where CloudFront stores website content. This could assist the website load faster by forwarding the content from the closest edge location.
- S3: This is to store all files needed for the website including images and HTML, CSS, and JavaScript code content of the website.
- DynamoDB: This function as a scalable NoSQL database that store the file such as image, video user information, file size and link to the file location like S3.

The explanation of the component usage

- Route 53: We used Route 53 to efficiently route user requests to AWS resources like CloudFront and Amplify Console to ensures fast delivery of website content, improving performance and user experience.
- Cloud Front: We used Cloud Front to spreads content worldwide, making the website load faster and improving user experience by reducing delays.
- Amplify Console: we used automatically updates the website files in the S3 bucket, so CloudFront always has the updates version to deliver the content.
- Edge Location: we implement the edge location to improve user experience by providing content from the nearest location.
- S3: We also used S3 to provide a secure, scalable, and highly available storage, ensuring the latest content is always accessible by web server.
- DynamoDB: is used in the diagram to manage and retrieve metadata for media files efficiently, keeping the system scalable, fast, and cost-effective as the application grows.

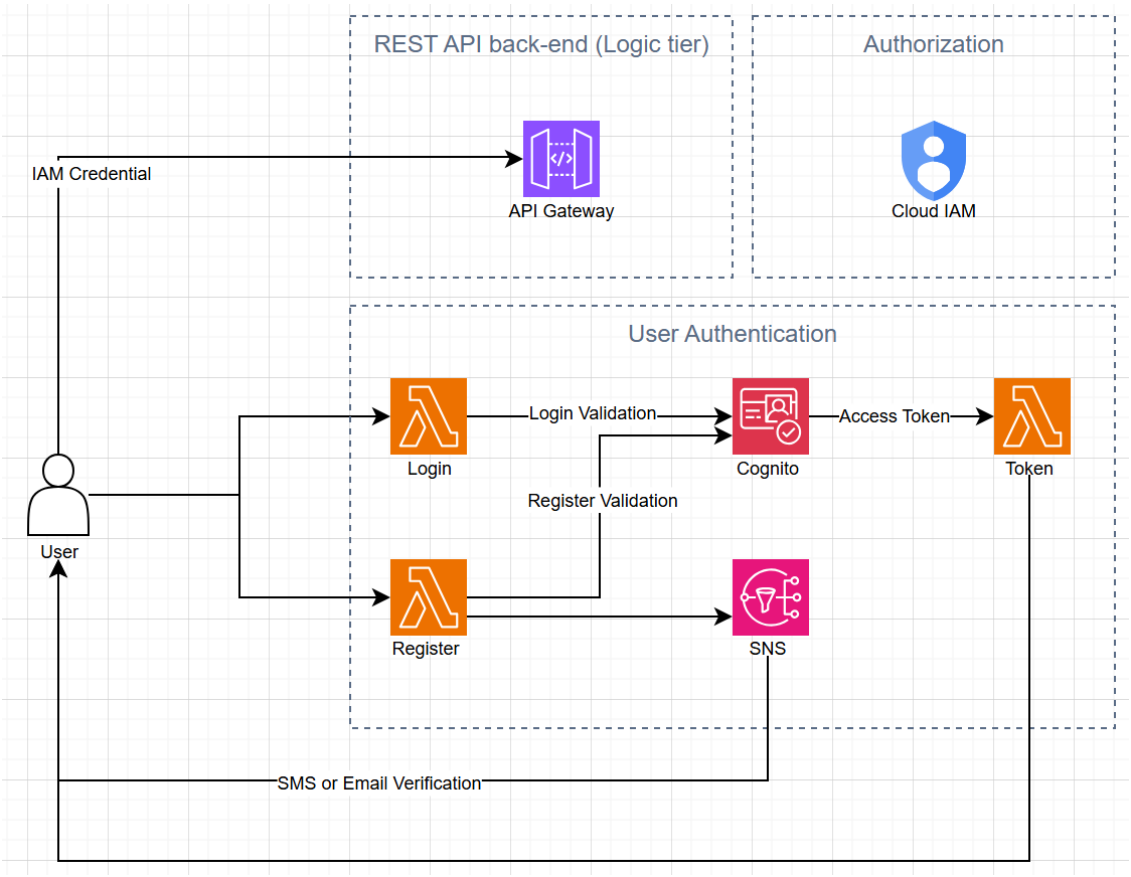


Figure 3 User Authentication

In this Figure, it shows the user authentication workflow and it starts when a user wants to request for a login or a registration via the API Gateway, which takes it into the backend process service. For the registration process, the Lambda function for registering looks through the details that user has input and proceeds to send a validation to the Cognito function to make a user profile in the user pool. And then, the SNS function sends a verification link or code through either SMS or Email so that the user can confirm their registration process which enhances the security. As for the login Lambda function, it validates the user credentials via Cognito then issues a JSON Web Token upon successful authentication. This token is used to authorize the user through IAM upon request, ensuring they have appropriate access to specific resources.

There are 5 component that contribute to Authentication tier:

The explanation of the component function

- API Gateway: simplifies building and managing APIs by providing a robust, scalable, and secure platform for routing and controlling API traffic. It integrates seamlessly with other AWS services like Lambda, EC2, and DynamoDB, making it ideal for building serverless applications, microservices, and more.
- Cloud IAM: a fundamental security service that helps control who is authenticated and authorized to use AWS resources. By managing identities, permissions, and roles, IAM plays a critical role in ensuring that AWS environments are secure and that resources are accessed appropriately by users and services.
- Lambda: AWS Lambda is a serverless computing service that eliminates the need for managing or provisioning servers. It dynamically adjusts resources to handle varying traffic loads and runs code in response to specific events or triggers. Lambda integrates smoothly with other AWS services, making it easy to execute functions based on predefined conditions.
- Cognito: Amazon Cognito is a comprehensive identity and access management (IAM) service that facilitates user authentication, permission management, and activity tracking. It offers multiple authentication methods, including username and password, social logins, and multi-factor authentication, to verify user identities securely.
- SNS (Simple Notification Service): Amazon SNS is a flexible and scalable messaging service designed for decoupling applications and services. It supports a wide variety of messaging patterns and allows you to send notifications across multiple channels and platforms efficiently.

  The explanation of the component usage

  - API Gateway usage: Our application backend relies on Lambda functions, while the API Gateway delivers multiple endpoints for user interaction. These endpoints cover functions such as media uploads, retrieving user information, and other key functions. By integrating with Cognito, API Gateway manages authentication through Cognito issuing JSON Web Token to users that has been authenticated ensuring safe and secure access.
  - Cloud IAM usage: IAM is the core of our security framework, making sure that only authorized users can get into our AWS material. It offers control based on the user's authority, strengthening the security of the photo album application.
  - Lambda usage: Lambda is essential to our serverless architecture, executing backend processes triggered by events like API Gateway requests and S3 updates.

Its automatic scaling ensures the system adapts efficiently to varying workload demands.

- Cognito usage: We use Cognito to handle authentication in our application by giving authorized users JSON Web Tokens, which has the user' s information used to validate each request made to the API Gateway.
- SNS usage: SNS is crucial in our architecture, facilitating communication and notifications between different components. It supports event-driven messaging, allowing various parts of the system to respond to specific events and stay synchronized.

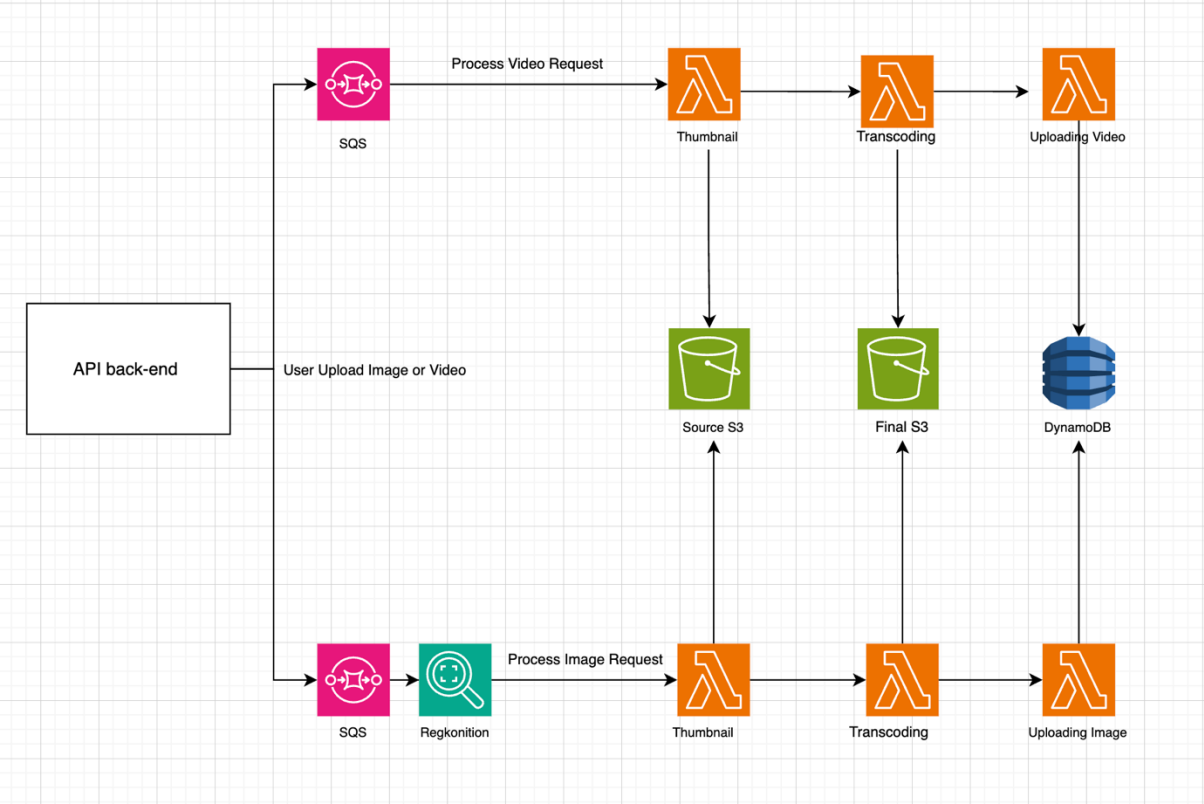


Figure 4 Video/Image Processing

Video processing

The video processing workflow begins when a user uploads any video to the website. Once uploaded it will be stored in the Source S3 Bucket, the video is placed in an SQS queue, which ensures that tasks are handled in the correct sequence. The first step in the process is the thumbnail Lambda function, which generates a small image preview to be displayed alongside the video for quick viewing. After that, the transcoding Lambda function retrieves the video from the queue and converts it into multiple formats, such as MP4, MOV, and WebM. It also adjusts the video resolution to ensure compatibility with different devices and

adapts the quality based on the user's internet speed, providing smooth experience for both high-speed and low-speed connections.

Once the video has been successfully transcoded and the thumbnail generated, the final step is uploading the processed video to a final S3 bucket for easy access and storage. In addition to this, important metadata about the video such as its format, resolution, and other relevant details are saved into DynamoDB. This metadata is crucial for situations where users may want to access specific versions of the video or retrieve information about the file.

By using SQS and Lambda functions, the system achieves a high level of scalability and reliability, ensuring that large volumes of videos are processed efficiently. If any issues occur during processing, the system automatically re-queues the video for retry, ensuring that no tasks are lost and maintaining the overall integrity of the workflow.

Image Processing

The image processing workflow begins when a user uploads a photo to the website. Once the photo is uploaded, it is stored in the Source S3 Bucket and placed in an SQS queue. This ensures that the uploaded images are handled sequentially and efficiently. The first step in the process is image recognition, where the system analyses the image to detect features such as objects, text, and faces. This analysis generates keywords that are stored in the image's metadata for easier categorization and retrieval.

Additionally, a thumbnail Lambda function generates a smaller version of the image, which is displayed on the interface without loading the full image, helping to reduce page load time. The Lambda resizing function adjusts the image's size, and while transcoding doesn't typically apply to images, if relevant, the image can be converted into different formats like JPG, PNG, or GIF to ensure compatibility across different devices and platforms.

Once the image has been analyzed, resized, and converted, it is uploaded to the final S3 Bucket for easy retrieval. Similar to video processing, the system also stores important metadata about the image, such as its format, resolution, and keywords, in DynamoDB, allowing users to easily access specific details or versions of the image.

By leveraging SQS, image recognition, and Lambda functions, the system ensures that image processing is scalable and reliable. If any issues occur during processing, the system automatically requeues the image, ensuring that no tasks are lost and that processing continues smoothly.

There are 6 components that contribute to Image/video processing:

- SNS (Simple Notification Service): Amazon SNS is a flexible and scalable messaging service designed for decoupling applications and services. It supports a wide variety

of messaging patterns and allows you to send notifications across multiple channels and platforms efficiently.

- SQS (Simple Queue Service): Amazon SQS is a message queueing service that enables asynchronous. It stores messages in a queue and ensures that they are processed sequentially. It also provides reliable message delivery, automatic retries, and error handling to ensure that tasks aren't lost.
- Regkonition: Amazon Rekognition is an image and video analysis service powered by machine learning that allows you to automatically detect objects, scenes, text, faces, and activities in images.
- Lambda: AWS Lambda is a serverless computing service that eliminates the need for managing or provisioning servers. It dynamically adjusts resources to handle varying traffic loads and runs code in response to specific events or triggers. Lambda integrates smoothly with other AWS services, making it easy to execute functions based on predefined conditions.
- S3: is used to store all files needed for the website including images and HTML, CSS, and JavaScript code content of the website.
- DynamoDB: is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability and is often used to store metadata about media files, such as formats, resolutions, processing status, and analysis results.

Explanation of the component usage

- SNS Usage: SNS is crucial in this process, it is use to trigger action by notifying when the image or video process are complete or any error during any stage of the process and ensure real time communication between service, it support delivery method include SMS, email and HTTP.
- SQS usage: SQS is essential in this design, because it is used to guarantee that any action will not be taken simultaneously, specifically for this part it prevents image or video that uploaded at the same time to be process simultaneously what might overwhelm the system by handling it in sequence and if any error occurs during the process, it will requeue and retry again.
- Regkonition usage: use of machine learning to analyze the images that were upload identifying objects, people, text or any detail in the image, in the purpose of creating an automatic tagging in the categorize each image properly where you can automatically flagging inappropriate image that is uploaded.

- Lambda usage: allows you to create build-in architecture that automatically responds to an event that occurs without manually managing the server itself. For this tier it triggers when user uploaded images or video It can scale and handle an unpredictable number of workloads automatically, which results in minimizing the usage of resource and more scalable and resilient.
- S3 usage: use for storing data that was uploaded into the website, in this part it handles large data files that were uploaded, ensures high durability, storage and security, its integration with other AWS services can enable smooth workflow.
- DynamoDB usage: use for storing metadata that related to media files include formatting, resolution, retrieval, search functionality and generate tag for images or video that were uploaded.

III. Design Rationale

A. Business Scenario Fulfilled and Justification

The proposed architecture fulfills the business scenario for the photo album application by providing AWS managed services to meet the company's requirements for scalability, efficiency, and security. Firstly, the solution ensures scalability using AWS Lambda, which allows for serverless, automatic scaling as user demand grows for future need, and eliminating the manual management. Moreover, to provide scalable we used Amazon S3 as a storage for media files, to integrate to work with other services to handle growing storage requirements. To enhance global performance and provide the best experience to the user, we implemented the combination of CloudFront and Route 53 working together to forward content from the closest edge location, this implementation reduces delay time and boosts the uploading speed for users around the world. Additionally, the upload process is designed to fulfill the business scenario by using Lambda functions for efficient media handling. When an image or video is uploaded to S3, Lambda functions automatically generate thumbnails, transcode videos, and manage uploads. These processed versions are then re-uploaded to S3. This automated, scalable setup ensures efficient media processing and aligns with the business need for a serverless solution. Furthermore, security is very important, which is why we use AWS Cognito to enhance security and provide users with multi-factor authentication, and IAM to manage permissions and keep resources secure. The architecture uses SNS and SQS to manage media processing tasks efficiently, keeping the system responsive. To reduce

costs, we chose DynamoDB, a NoSQL database, to replace the traditional database, which incurs higher costs, and to better handle growing data volumes in the future. Overall, this proposal aims to provide the best experience for user with less cost, ensures the system remains scalable and secure for current business needs while being adaptable for future innovate with serverless technologies.

B. Alternative Solutions Comparison

I. SQL vs NoSQL database

- SQL databases define the structure of data in tables or columns. They ensure consistency and reliability, making them ideal for applications requiring strong data integrity. On the other hand, NoSQL databases are designed for scalability and flexibility, supporting a variety of data models that can distribute data across multiple end devices easily. NoSQL is optimized for fast-paced environments.

II. 2tier vs 3-tier

- 2-tier architectural design consists of two main layers: the user interface and the database. This means clients interact directly with the database for any data retrieval or storage through database access methods or an Application Programming Interface (API). This setup is easy to design and manage, has faster response times between the two layers, and is cost-efficient. On the other hand, a 3-tier architectural design, as used in your architecture, improves security and access control before clients can reach the data. It also manages traffic more efficiently across the database, improving performance and scalability to handle high traffic without overload.

III. Virtual machine vs serverless

- A Virtual Machine (VM) is a software-based emulation of a physical server. With VMs, you have control and the ability to configure custom settings or install any software, but VMs require more resources, utilizing full CPU and memory. Serverless computing, on the other hand, allows you to run code without provisioning or managing the server. This means everything is handled by the cloud provider. It is also cost-efficient and automatically scales up or down based on demand.

C. Design Criteria

1. Performance

The architecture leverages AWS Lambda, which executes functions in response to events (such as API requests, SNS notifications, and S3 events). This ensures that only the necessary code is executed on demand, keeping performance optimized for various workloads. AWS S3 and DynamoDB are used for efficient, high-performance data storage and retrieval.

2. Scalability

Lambda functions, API Gateway, SNS, and S3 are all designed to scale automatically based on incoming traffic. The architecture's use of SQS queues for video and image processing ensures asynchronous processing, enabling the system to handle varying workloads without overloading backend services.

3. Reliability

S3, DynamoDB, Lambda, and SQS are built for high availability and fault tolerance, ensuring the application remains reliable even in the event of hardware or network failures. SQS and SNS provide mechanisms for reliable message delivery, ensuring that no events are lost even during high traffic or intermittent failures.

4. Security

Cognito is used for user authentication, generating JWT tokens to verify users and ensuring secure access to API Gateway. This ensures that only authenticated users can access the backend services. IAM roles and policies ensure that each component of the system only has access to the resources it needs, implementing the principle of least privilege. This ensures that even if one service is compromised, it cannot affect other services. And with API Gateway, IAM policies manage who can access specific API endpoints, further securing the backend logic. Next, our data is securely stored in S3 and DynamoDB with encryption at rest and in transit.

The AWS services selected offer an optimal mix of performance, scalability, dependability, and security, perfectly suited to meet the design requirements of the Photo Album application. These services work together to form an architecture capable of managing fluctuating workloads, safeguarding data, and delivering seamless, secure experience for users.

5. Cost

The configuration for AWS Cognito is based on an estimate of 750 monthly active users (MAUs), which assumes a user base of 500 daily active users. We also configured **5% of**

users to use federated identities, which is typical for modern web applications offering multiple authentication options. Given the free tier of Cognito, which covers up to 50,000 MAUs and 50 federated identities, the current setup incurs no cost for authentication services.

The AWS S3 setup includes 1 TB of storage and 1 million GET/PUT requests per month, supporting 500 daily users uploading and downloading media. This configuration provides ample capacity for storing and accessing photos and videos, with a monthly cost of $23.40 for storage and request handling. It's scalable, allowing for growth as media needs increase.

AWS Lambda is configured with 100,000 monthly invocations, 512 MB of memory, and a 3-second average execution time per invocation. It handles tasks like media processing, transcoding, and thumbnail generation, playing a key role in the application's event-driven architecture. The estimated cost is $2.52 per month, offering a cost-effective solution with sufficient computing power for these backend tasks without requiring dedicated servers.

API Gateway is set to handle 1 million requests per month, accommodating the expected activity of 500 daily users uploading, retrieving, and managing media. The cost is $4.79 per month, providing a secure and scalable solution for managing all API requests efficiently.

AWS SQS is configured to handle 500,000 messages per month, managing asynchronous tasks like media uploads, processing, and system notifications. As a crucial part of the serverless and event-driven architecture, it enables decoupled task processing. This setup is cost-efficient, with a monthly cost of $0.

The AWS Amplify configuration includes 600 build minutes and 10 GB of storage per month, supporting continuous deployment of the Photo Album app's frontend. With two daily builds averaging 10 minutes each, it ensures seamless updates without downtime. The total monthly cost is $6.23, covering both build time and storage, offering an affordable solution for hosting and deployment.

AWS CloudFront is configured to manage 2 TB of data transfer to the internet, 2 TB from origin, and 1 million requests per month. This setup ensures fast global delivery of media files, reducing latency for users, especially those outside Australia. The monthly cost is $211,

covering data transfer and request handling. CloudFront is crucial for delivering content quickly worldwide, enhancing the user experience.

DynamoDB is configured with 50 GB of storage and 500,000 read/write requests per month, storing metadata like tags, descriptions, and filenames for uploaded media. It ensures fast, reliable access as users interact with the application. The estimated monthly cost is $25.20, covering both storage and database activity for an average item size of 15 KB, offering a scalable and efficient solution for metadata management.

AWS SNS is configured to manage 1 million notifications per month, handling system alerts related to media uploads, user interactions, and processing updates. With a monthly cost of $0, it is a cost-effective solution for managing notifications across the application.

Amazon Rekognition is configured to process 5,000 images and 100 minutes of video analysis per month, meeting the application's needs for media tagging, object detection, and AI-based analysis. It helps automate metadata generation and improves the user experience with features like image search and video tagging. The total monthly cost is $19, making it a cost-effective solution for these advanced capabilities.

Lastly, Amazon Route 53 is configured with 1 hosted zone, 1 million DNS queries per month, and 1,000 IP (CIDR) blocks. This setup ensures fast and reliable domain name resolution, along with advanced routing features like geolocation and IP-based routing. These capabilities are vital for maintaining high availability and low-latency access, optimizing user experience across different regions. The estimated monthly cost is $1.10, providing a cost-effective solution for DNS management with advanced routing.

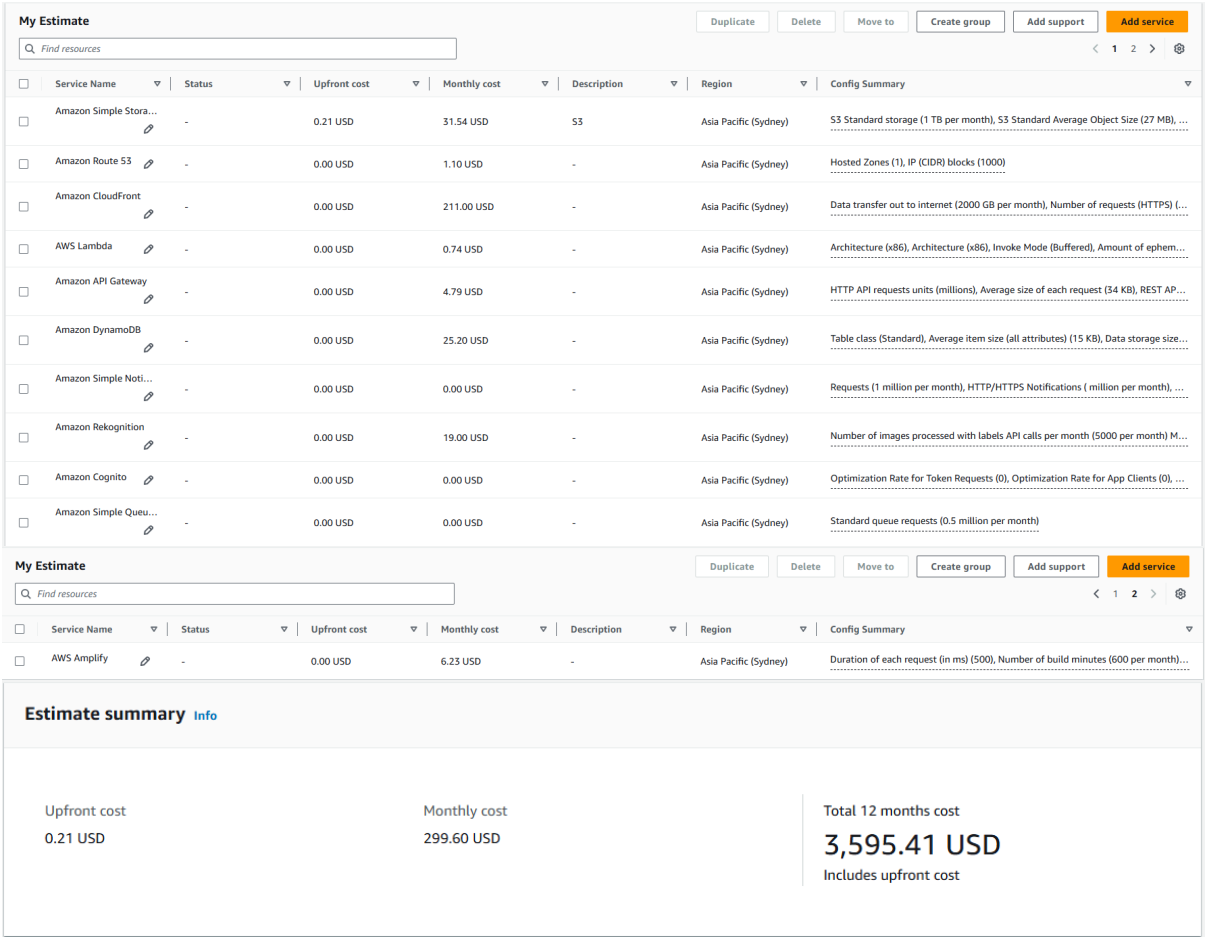


Figure 5 Cost Estimation

IV. Conclusion

In conclusion, the architecture designed for the photo album application provides a scalable solution that effectively meets the business requirements in terms of good performance, efficiency, and cost management. Utilizing a combination of AWS managed services, such as S3, CloudFront, Route 53, DynamoDB, Lambda, and Cognito, the system ensures efficient content delivery, secure user management, and automated media processing. The serverless and event-driven approach guarantees that the application can automatically scale with increasing user demand, minimizing operational overhead while maintaining cost-effectiveness.

Additionally, the architecture's modular and flexible design supports future growth and the addition of new features, such as enhanced media processing capabilities. This adaptability ensures that the platform remains robust, secure, and responsive as the application evolves. Overall, this solution is strategically built to optimize performance, reduce costs, and maintain high security standards, positioning the application for long term success and scalability.

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